

Behavioural Research in Statistical Methods Project Report - 2

Team Name: Earphones

Experiment: Mnemonic Similarity Task (MST)

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In this phase, we implemented confirmatory models centered on correctness and response speed.

We also tested additional variables beyond REC and LDI, including response-category structure and lure-bin effects.

1. Introduction

Phase 1 established a clean, reproducible base and suggested a post-boundary cost for some memory outcomes. In Phase 2, our aim was stricter: test correctness and response speed directly at the test-trial level, include additional predictors, and enforce explicit correction for multiple testing.

This phase followed class guidance on variable-type aware testing, assumption checks, outlier diagnostics, and effect-size oriented reporting. We retained continuity with the MST literature and event-segmentation framing (Zacks and Swallow, 2007) (Swallow et al., 2009) (Stark et al., 2019) (Yassa and Stark, 2011) (Morse et al., 2023).

1.1. Phase 2 questions

1. Does boundary position affect test correctness (`correct` = 0/1) when modeled directly?
2. Does boundary position affect test response speed (`response_rt`) after accounting for item role and condition?
3. Do response-category patterns (`old/new/similar`) differ by boundary position?
4. Do lure-bin difficulty and stimulus class explain additional variance?
5. Do speed-accuracy summaries support meaningful differences between correct and incorrect responses?

CONDITION	PAIRED n	TEST TRIALS PER PARTICIPANT	TOTAL ANALYZED TEST TRIALS
Item Shift Only	56	150	8400
Item + Task Shift	49	150	7350
Task Shift Only	53	150	7950

Table 1: Phase 2 sample overview derived from paired participants and test-trial data used in confirmatory modeling.

2. Methods

2.1. Data and preprocessing

We reused the paired dataset from Phase 1 and operated primarily on test trials. Key analysis columns were correctness (`correct`), response label (`response_label`), response time (`response_rt`), condition, boundary position, item role, lure bin, and stimulus class.

Quality checks included missingness summaries, RT artifact flags, participant-level RT outlier checks, and normality diagnostics of participant mean log RT. We observed no missingness in core test-trial fields and only minimal missingness in a participant-level covariate (`encoding_task_accuracy` , 2 participants).

2.2. Variables and measures

Phase 2 primary outcomes were:

1. Test correctness (`correct`)
2. Log-transformed response speed (log of `response_rt`)

Secondary outcomes were:

1. Response-category profile (old/new/similar)
2. Lure-specific similar-response probability as a function of lure bin
3. Speed-accuracy summary: mean correct RT, mean incorrect RT, and their difference

For continuity, REC and LDI remained available as bridge metrics: $REC = P(\text{old } m \text{ id Target}) - P(\text{old } m \text{ id Foil})$ $LDI = P(\text{similar } m \text{ id Lure}) - P(\text{similar } m \text{ id Foil})$

2.3. Statistical strategy

Primary correctness was modeled with clustered logistic GEE (participant-level clustering, exchangeable working correlation). Primary RT was modeled on log RT with participant-aware modeling; due singularity in the richest mixed specification, a robust Gaussian GEE fallback model was used and reported.

Secondary analyses included chi-square tests for response-category profile changes and a lure-bin GEE model for similar responses on lure trials. Multiple comparisons were corrected within analysis families using Holm, with BH/FDR also reported in outputs.

2.4. Precision and power context

To make null and weak effects interpretable, we added a precision-context summary by condition. For proportion-like outcomes, the worst-case 95% CI half-width at $p = 0.5$ was approximately 0.131 (item_only), 0.140 (both), and 0.135 (task_only). For participant-mean RT, 95% CI half-widths were approximately 0.308s (item_only), 0.240s (both), and 0.153s (task_only).

These values provide a practical interpretation layer: effects smaller than these ranges are harder to distinguish cleanly without larger samples, especially for proportion endpoints.

3. Results

3.1. Quality-control checks

Core test-phase variables had no missingness. RT artifact counts were low relative to dataset size (1 very fast trial $< 0.2s$; 23 very slow trials $> 30s$). Participant-level mean log RT departed from perfect normality (Shapiro-Wilk $W = 0.953$, $p < .001$), so robust model choices and fallback logic were retained.

Additional RT diagnostics using an OLS approximation for residual checks showed non-normal residual structure (Shapiro-Wilk $W = 0.949$, $p < .001$) and a weak absolute-residual versus fitted correlation (about 0.086), consistent with mild heteroscedasticity. This further justified reporting the robust clustered RT model and robustness reruns.

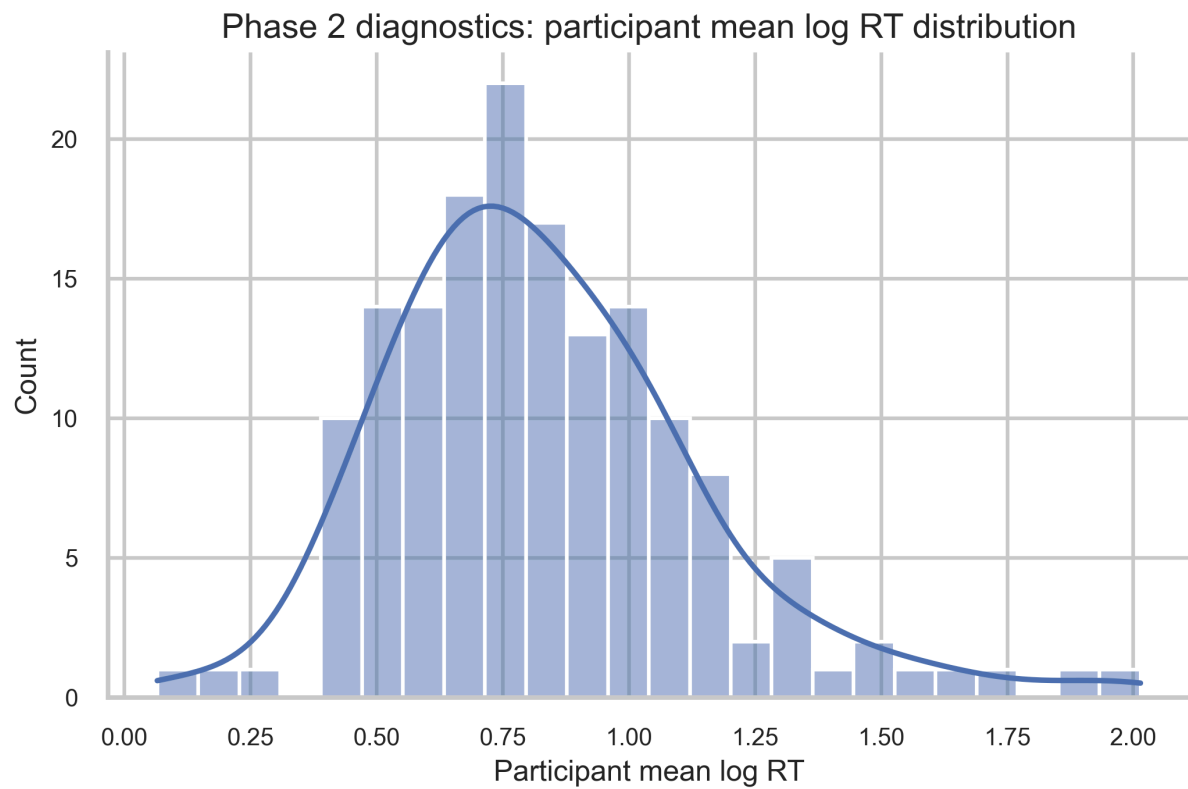


Figure 1: Diagnostic histogram of participant-level mean log RT used for distributional assessment.

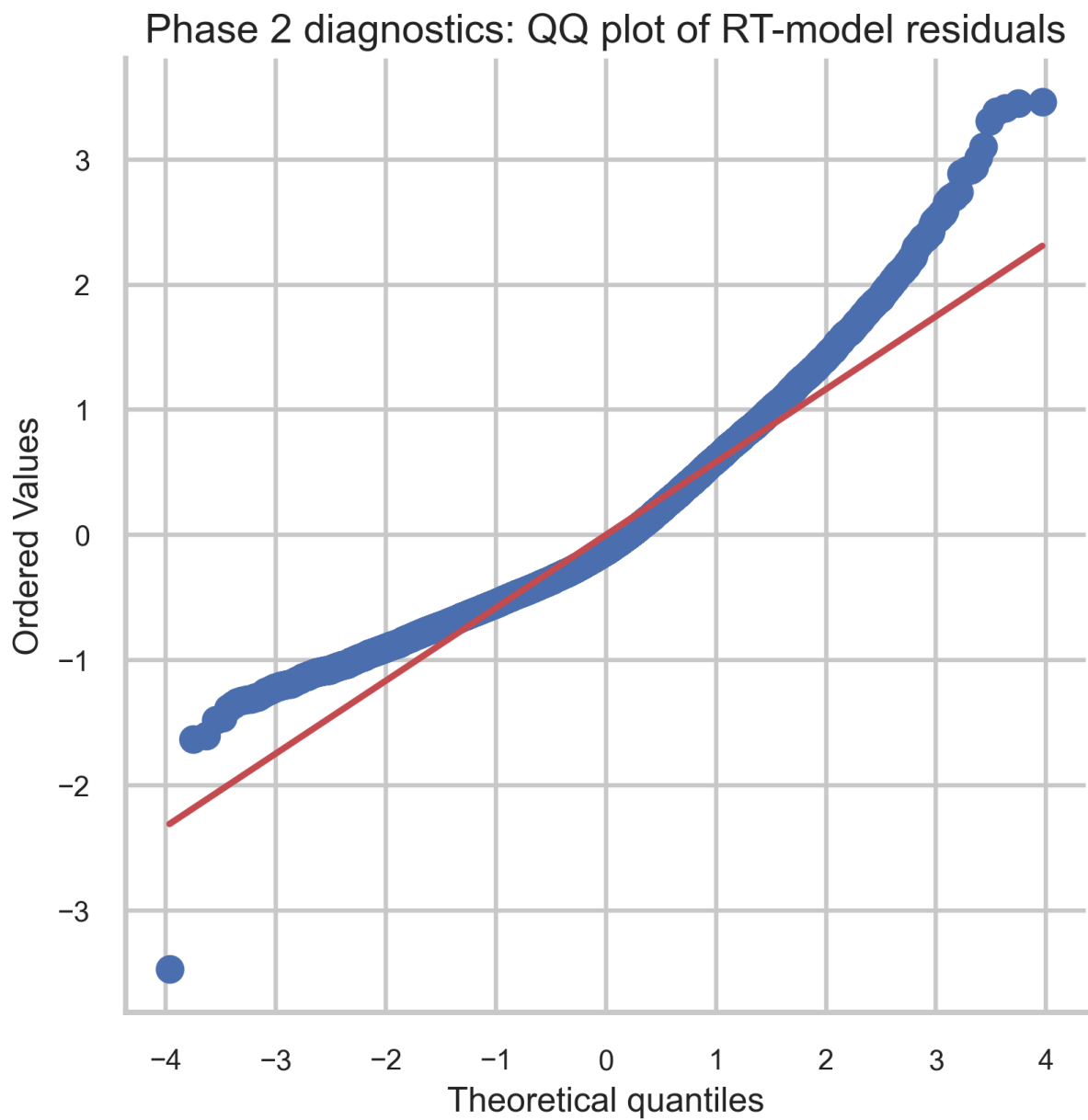


Figure 2: QQ plot of RT-model residuals (diagnostic approximation), showing departures from strict normality.

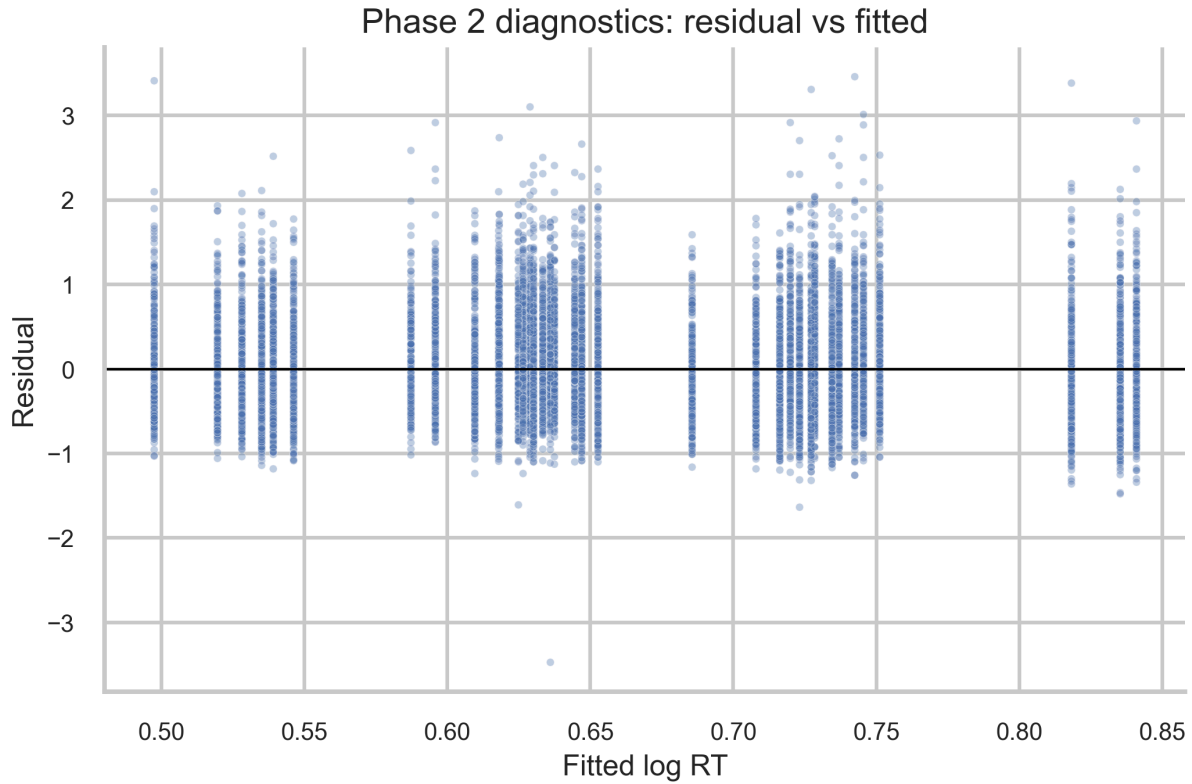


Figure 3: Residual-versus-fitted diagnostic for RT model approximation, indicating only mild structure in residual spread.

3.2. Primary model 1: Correct/incorrect responses

In the confirmatory correctness model, broad boundary-position main effects were not the dominant signal after adjustment. Instead, correctness varied strongly with item role and stimulus structure.

Key coefficients (uncorrected model-level terms):

1. lure vs target : OR = 0.58, $p < .001$
2. Scenes vs Objects : OR = 0.66, $p < .001$
3. Lure-bin centered slope: OR = 1.09, $p < .001$
4. both vs item_only : OR = 0.80, $p = .021$

After Holm correction, the strongest retained terms were item role, stimulus class, and lure-bin effect.

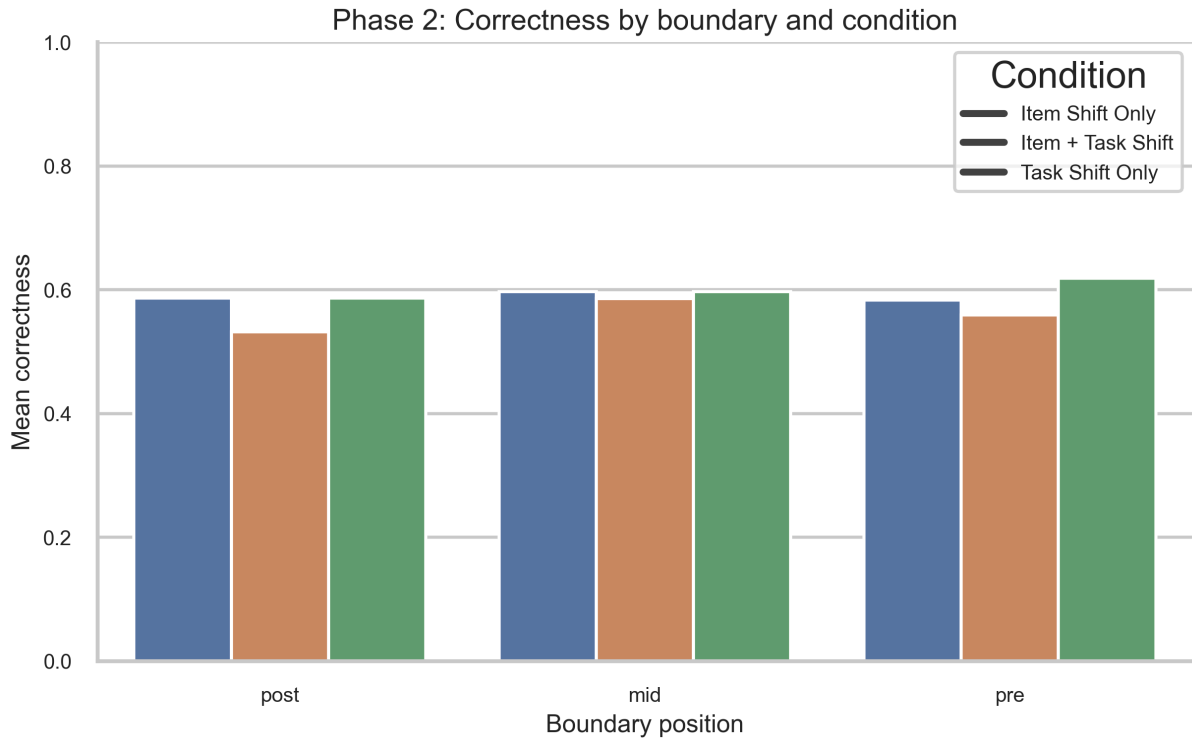


Figure 4: Phase 2 correctness by boundary and condition. The boundary pattern is present descriptively but weaker than item-role and lure-difficulty effects in adjusted modeling.

3.3. Primary model 2: Response speed

For response speed, the final fitted primary model used the Gaussian GEE fallback on log RT. The major robust effects were:

1. Correct responses were faster than incorrect responses ($\exp(\beta) = 0.922$, $p < .001$).
2. Lure trials were slower than target trials ($\exp(\beta) = 1.105$, $p < .001$).
3. Task Shift Only condition showed faster RT than Item Shift Only in adjusted comparison ($\exp(\beta) = 0.880$, $p = .013$).

Boundary main effects were not the strongest adjusted terms.

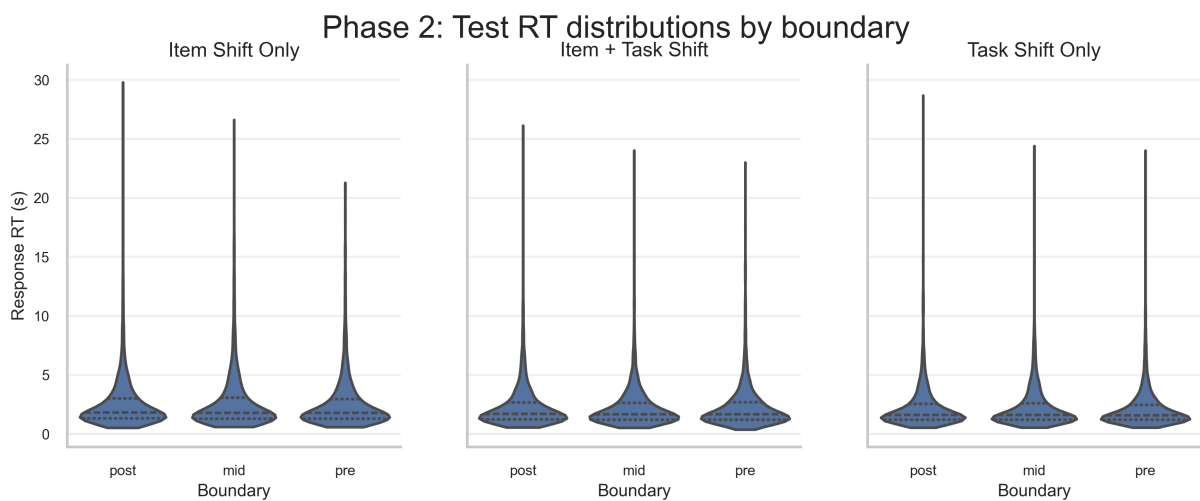


Figure 5: Phase 2 response-time distributions by boundary and condition. Distribution-aware plots were used instead of mean-only summaries.

3.4. Secondary analyses beyond REC and LDI

Response-category structure (chi-square tests) showed a significant boundary-linked shift only for target trials in the Item + Task Shift condition: $\chi^2(4) = 22.55$, $p < .001$, Cramer's $V = 0.062$.

In the lure-focused model, similar-response probability increased with lure-bin level ($OR = 1.17$, $p < .001$), indicating stronger discrimination as lure similarity reduced. Scene stimuli showed lower similar-response probability than objects ($OR = 0.68$, $p < .001$).

The speed-accuracy summary also showed a consistent pattern: incorrect responses were slower on average than correct responses (overall mean difference about 0.278 s).

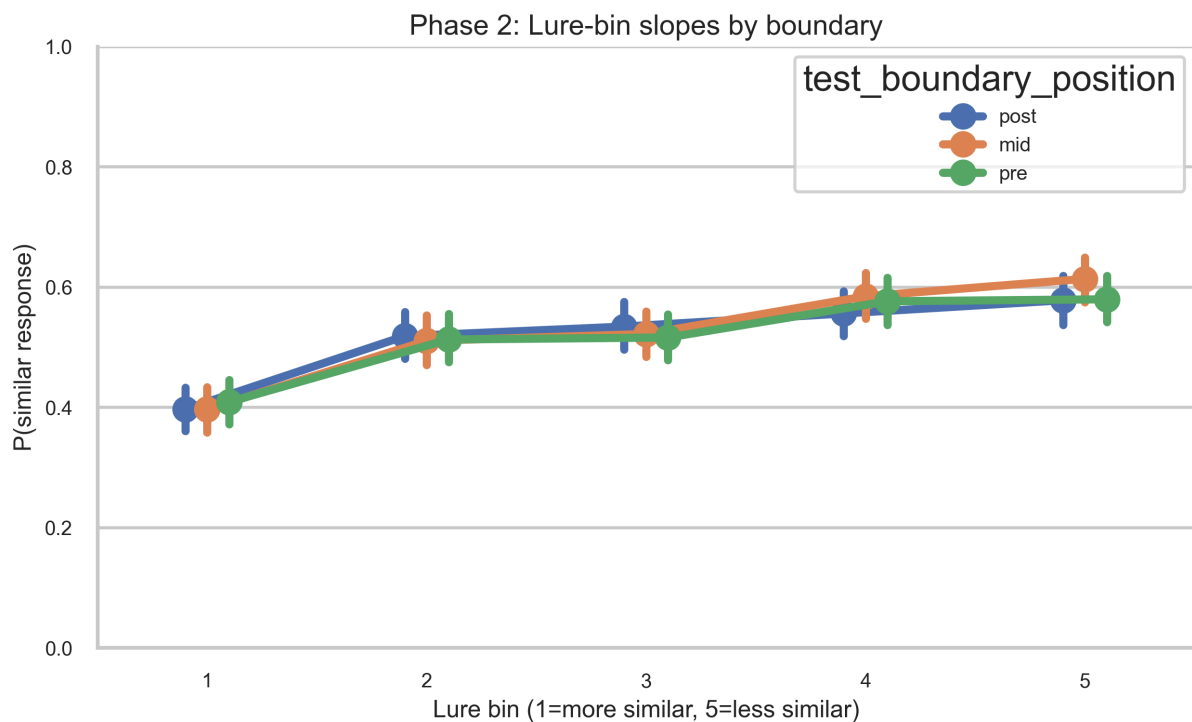


Figure 6: Phase 2 lure-bin slopes by boundary position. Similar-response probability increases with lure bin, indicating expected MST difficulty behavior.

3.5. Integrative interpretation

Phase 2 did not show a dominant omnibus boundary main effect across all adjusted models. Instead, the main robust structure came from item role, stimulus class, lure difficulty, and speed-accuracy coupling. This sharpens Phase 1 interpretation: boundary effects are present but conditional, and should be interpreted alongside stimulus and response-process variables rather than in isolation.

3.6. Robustness analyses

The high-impact robustness package was added in three parts.

1. **Outlier-trimmed RT rerun:** Excluding participant-level RT outliers did not materially change the RT story. Correct responses remained faster than incorrect responses, and lure trials remained slower than target trials.

2. **Split correctness models:** Modeling target and lure correctness separately preserved key patterns. For lure trials, lure-bin and stimulus-class effects remained strong. For target trials, the strongest adjusted effects remained condition/stimulus dependent rather than a uniform boundary main effect.
3. **Focused boundary contrasts:** In corrected contrast testing, the clearest retained boundary effect was in Item + Task Shift target trials for post-minus-mid correctness (Holm-corrected significance retained).

Overall, the confirmatory conclusions were stable under these robustness checks.

4. Conclusion

Phase 2 achieved the confirmatory objectives and expanded coverage beyond REC and LDI by modeling correctness, response speed, and response categories directly.

Main takeaways:

1. Correctness and RT analyses are now trial-level and model-based.
2. Strongest robust effects are item-role and lure-difficulty related.
3. Speed-accuracy coupling is meaningful (incorrect responses slower on average).
4. Boundary-related effects appear context-dependent rather than uniformly dominant.

This phase completes the methodological progression from exploratory to confirmatory analysis while aligning with class expectations for assumptions, categorical testing, multiple-comparison control, and effect-size reporting.

5. Codebase and contributions

The source code for this project is available at <https://github.com/ch-pavan/brim>^{oo}.

This phase was completed collaboratively:

1. **Sambu Aneesh:** Framed confirmatory questions and statistical interpretation logic.
2. **Renu Sree Vyshnavi:** Led results interpretation and quality-control narrative.
3. **Pavan Harshit:** Implemented Phase 2 analysis pipeline, generated outputs, and integrated report assets.

References

- [1] J. M. Zacks and K. M. Swallow, "Event Segmentation," **Current Directions in Psychological Science**, 16(2), 80-84, 2007.
- [2] K. M. Swallow, J. M. Zacks, and R. A. Abrams, "Event Boundaries in Perception Affect Memory Encoding and Updating," **Journal of Experimental Psychology: General**, 138(2), 236-257, 2009.
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- [4] M. A. Yassa and C. E. L. Stark, "Pattern Separation in the Hippocampus," **Trends in Neurosciences**, 34(10), 515-525, 2011.
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